

Optimized code:-

$t = a + b$

$q = t + t$

- Removed redundant computation $y = b + a$
- Removed identity operations $(*1)$.
- Reduced number of statements.

13) $a = 5$

$b = a + 2$

if $(b > 5)$ goto 6

$a = 10$

goto 7

$b = a + 1$

$c = b + a$

We label each definition

- $d1: a = 5$ (line 1)
- $d2: b = a + 2$ (line 2)
- $d3: a = 10$ (line 4)
- $d4: b = a + 1$ (line 6)

Control flow:-

- $1 \rightarrow 2 \rightarrow 3$
- $3 \rightarrow 4 \rightarrow 5 \rightarrow 7$
- $3 \rightarrow 6 \rightarrow 7$

Line	Statement	GEN	KILL
1	$a = 5$	$\{d1\}$	$\{d3\}$
2	$b = a + 2$	$\{d2\}$	$\{d4\}$
3	if	ϕ	ϕ
4	$a = 10$	$\{d3\}$	$\{d1\}$
5	goto	ϕ	ϕ